

MH1301 Discrete Mathematics

Handout 1 - Basics of Counting

We begin with Part I of the course on counting. In this set of slides, we cover the basic counting principles, namely the sum and product rules. Then we also cover the inclusion-exclusion principle, which allows us to count the number of elements in a union of sets.

- Product Rule.
- Sum Rule.
- Inclusion-Exclusion Principle.

Recall from MH1300

Definition

A **set** is an **unordered** collection of objects, called elements.
Two sets are equal if and only if they have exactly the same elements.

Notation: We write $a \in A$ to denote a is an element of the set A and likewise, $a \notin A$ means a is not an element of A .

The **axiom of extension** says that

- a set is completely determined by its elements,
- the order in which the elements are listed is irrelevant,
- some elements may be listed more than once.

Example.

$$\{a, b, c\} = \{b, c, a\} = \{c, a, b\} = \{a, b, c, a\}$$

Recall from MH1300

Notations

- We normally use uppercase letters to denote sets, e.g., A , B , etc.
- A set is said to **contain** its elements.
- An element x is inside set A is denote by $x \in A$.
- We also use $A \ni x$ for set A contains element x .
- Similarly, we use $x \notin A$ to mean element x is *not contained* in set A .
- A way to define a set is to indicate its elements between a pair of braces, i.e. Set-Roster notation.

E.g. $\{1, 2, 3\}$, $\{1, 2, 3, \dots, 50\}$, $\{5, 10, 15, 20, \dots\}$.

- The symbol $\dots$ is called an **ellipsis** and is read “and so forth”.

Recall from MH1300

<i>Symbol</i>	<i>Set</i>
$\mathbb{N}$	<i>set of natural numbers</i>
$\mathbb{Z}$	<i>set of integers</i>
$\mathbb{Q}$	<i>set of rational numbers, or quotients of integers</i>
$\mathbb{R}$	<i>set of real numbers</i>
$\mathbb{C}$	<i>set of complex numbers</i>

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$
→ includes zero
- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- $\mathbb{Z}^{\text{nonneg}} = \mathbb{N}$

Recall from MH1300

Definition

Let S denote a set and let $P(x)$ be a property that elements of S may or may not satisfy. We may define a new set to be **the set of all elements x in S such that $P(x)$ is true.**

Denote by

$$\{x \in S \mid P(x)\}$$

the set of all such that

which is read “the set of all x in S such that $P(x)$.”

Example.

$$\mathbb{Z}^+ = \{x \in \mathbb{Z} \mid x > 0\}$$

“The set of all integers x such that $x > 0$.”

Recall from MH1300

Definition

If A and B are sets, then A is called a **subset** of B , written $A \subseteq B$ if and only if every element of A is also an element of B .

Symbolically:

$A \subseteq B$ means that For all elements x , if $x \in A$ then $x \in B$.

We also use the phrases A is contained in B and B contains A .

$A \not\subseteq B$ means that There is an element x such that $x \in A$ and $x \notin B$.

Definition

Let A and B be sets. A is a **proper subset** of B , written $A \subsetneq B$, if and only if every element of A is in B but there is at least one element of B that is not in A .

Cardinality

Definition

The **size** or **cardinality** of a set S , denoted by $|S|$, is the number of elements in S .

Counting is the task of finding the cardinality of a given set.

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Example. What is the size of $\{a, b, c, a\}$? 3

Recall from MH1300

Definition

A **sequence** is an **ordered** collection of elements (components/entries), not necessarily distinct.

Example

- (a, b, c)
- (b, a, c)
- (a, b, a)

Are they the same? *No*

Are they sets? *In theory, yes*

Example - Dice

Say you have a six-sided die and a two-sided coin (head (H) on one side and tail (T) on the other).

What is the number of possible outcomes when both the die and the coin are tossed?

Solution: All possible outcomes are:

$$\{(1, H), (2, H), (3, H), (4, H), (5, H), (6, H), \\ (1, T), (2, T), (3, T), (4, T), (5, T), (6, T)\}.$$

There are 6 possible outcomes for the die and 2 for the coin, so the total number of outcomes is $6 \times 2 = 12$.

Product Rule



Product Rule

Given a set S of sequences of length k , where

- There are n_1 possible first entries.
- There are n_2 possible second entries for each first entry.
- There are n_3 possible third entries for each combination of first and second entries, and so on up to n_k .

the size of S is $n_1 \cdot n_2 \cdot \dots \cdot n_k$.

Example - License Plates

Example 5 (Section 6.1)

How many different license plates are available if each plate contains a sequence of 3 letters followed by 3 digits?

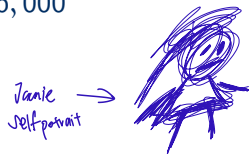
26

10

Solution: There are 26 choices for each of the three letters and 10 choices for each of the three digits. By product rule, there are a total of

$$26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 17,576,000$$

possible license plates.



Example - Permutation

Permutation

A permutation of a set S is a sequence that contains every element of S exactly once.

For example, the permutations of the set $S = \{1, 2, 3\}$ are the sequences $(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)$.

Example - Permutation

How many permutations does an n element set have?

Solution:

- The first entry in the permutation sequence can be chosen in n possible ways.
- Each such choice leaves out $n - 1$ possibilities for the second element.
- Each combination of choices for the first two elements leaves out $n - 2$ combinations for the third element.
- Continuing like this, we get that

The number of permutations of an n element set is

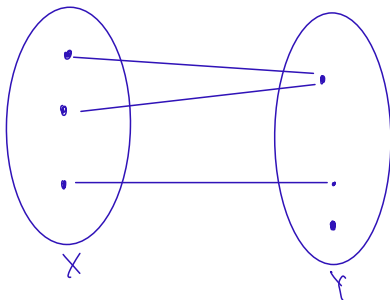
$$n \cdot (n - 1) \cdot (n - 2) \cdots 1 = n!.$$

Functions

Defintion

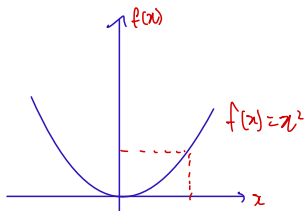
Let X and Y be sets. A function $f : X \rightarrow Y$ is a relation between two sets X and Y such that for **every** element x in X ,
there is **exactly** one element y in Y which x is mapped to.

We write $f(x) = y$.

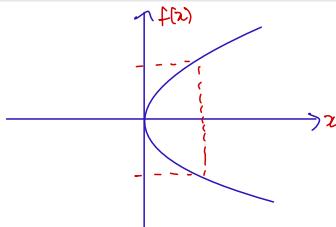


Functions

Question: Is $f(x) = x^2$ a function? *Yes*



Question: How about $f(x) = \pm\sqrt{x}$? *No*



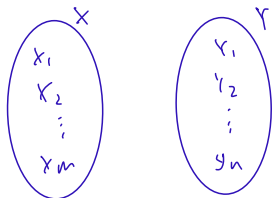
Functions

Example 6 (Section 6.1)

How many functions are there from a set X with m elements to set Y with n elements?

Solution: A function corresponds to a choice of one of the n elements in Y for each of the m elements in X .

By product rule, there are $\overbrace{n \cdot n \cdots n}^m = n^m$ such functions.

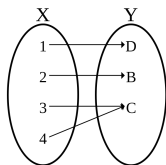


Be careful, not m^n !

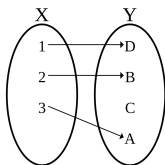
Functions

Definition: Injection, Surjection, Bijection

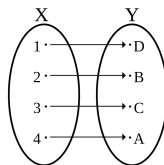
- A function $f : X \rightarrow Y$ is a **surjective** function if every element in Y is mapped from **at least** one element in X .
- A function $f : X \rightarrow Y$ is an **injective** function if every element in Y is mapped from **at most** one element in X .
- A function $f : X \rightarrow Y$ is a **bijective** function if every element in Y is mapped from **exactly** one element in X .



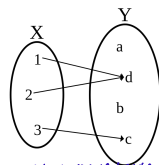
Surjection



Injection



Bijection



not injective Not surjective
not bijective

↳ but still a function

bijective $\iff$ surjective + injective

Mapping Rules

Mapping Rules

Assume X and Y are finite sets.

- A function $f : X \rightarrow Y$ is **surjective** $\implies |X| \geq |Y|$.
- A function $f : X \rightarrow Y$ is **injective** $\implies |X| \leq |Y|$.
- A function $f : X \rightarrow Y$ is **bijective** $\implies |X| = |Y|$.

Bijections are very useful in counting.

How to count $|Y|$? Find a bijection $f : X \rightarrow Y$, count $|X|$.

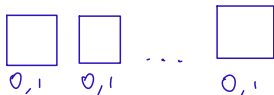
- Do not overcount or undercount!

Example - Binary Strings

Question

How many n bit binary sequences are there?

Solution: Each entry of the sequence has two possibilities: 0 or 1. By product rule, there are in total $2 \cdot 2 \cdots 2 = 2^n$ such sequences.



$\varphi: \{\text{all binary seq. of length } n\} \rightarrow \overset{\text{power set}}{\mathcal{P}(S)}$ Set of all subsets of S .

Example - Subsets $\varphi((b_1, b_2, b_3, \dots, b_n)) = T \subseteq S$
where $i \in T$ iff $b_i = 1$

How many different subsets of a finite set $S = \{1, \dots, n\}$ are there?

Solution: We construct a bijective function from the subsets of S to the set of n bit binary sequences: the subset $\{s_1, s_2, \dots, s_k\}$ maps to the bit sequence that has ones in positions $s_1, \dots, s_k$ and zeros in all the other positions.

For example, if $n = 7$, the subset $\{3, 4, 6\}$ maps to the bit sequence $(0, 0, 1, 1, 0, 1, 0)$.

$\downarrow$ $\downarrow$ $\downarrow$
 b_3 b_4 b_6

Clearly every subset of S maps to some n bit sequence. It is also true that every n bit sequence is mapped by a subset: The sequence $(b_1, \dots, b_n)$ is mapped by the set of all i between 1 and n such that $b_i = 1$.

So this is a bijection. Because we already know that the number of n bit sequences is 2^n , by the mapping rule, the number of subsets of S must also be 2^n .

Example - Textbooks

Suppose there are 26 MH1301 textbooks in NTU library and 15 in NIE library. How many textbooks are there in total?

Solution: $26 + 15 = 41$.

Sum Rule

Sum Rule

If $A_1, A_2, \dots, A_k$ are **disjoint** sets, then

$$|A_1 \cup \dots \cup A_k| = |A_1| + \dots + |A_k|.$$

Example - Passwords

Passwords on a computer are of six to eight characters long, where each character is either an uppercase letter or a digit. How many possible passwords are there?

26 10

Solution: Let X be the total number of possible passwords, and X_6, X_7, X_8 denote the number of possible passwords of length 6, 7, 8, respectively. By the sum rule, $X = X_6 + X_7 + X_8$.

By the product rule,

$$X_6 = (26 + 10)^6 = 36^6,$$

$$X_7 = (26 + 10)^7 = 36^7,$$

$$X_8 = (26 + 10)^8 = 36^8.$$

So the total number of possible passwords is

$$36^6 + 36^7 + 36^8 = 2,901,650,853,888 \approx 2.9 \cdot 10^{12}$$

Example - Strings

Example 18 (Section 6.1)

How many 0-1 bit strings of length 8 either start with a '1' bit or end with the two bits '00'?

Solution:

- Let A_1 be the set of strings start with '1'. By the product rule,

$$|A_1| = 2^7 = 128$$

- Let A_2 be the set of strings end with '00'. By the product rule,

WRONG

$$|A_2| = 2^6 = 64$$

- By the sum rule, the final answer is $128 + 64 = 192$.

The two sets are not disjoint. We **overcounted!**

A string can both start with '1' and end with '00'.

Inclusion-Exclusion Principle

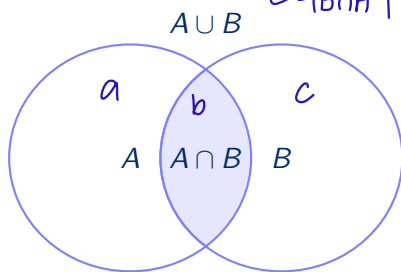
Inclusion-Exclusion Principle for two sets

Given two sets A_1 and A_2 , not necessarily disjoint, the number of elements in the union of these sets is

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|.$$

$$a+b+c = a+b+b+c-b$$

$$c = |B \cap A^c|$$



Venn diagram

$$|A_1| + |A_2| - |A_1 \cap A_2|$$

Note that if A_1 and A_2 are disjoint, then $|A_1 \cap A_2| = 0$, which implies the sum rule.

Example - Strings

Example 18 (Section 6.1)

How many 0-1 bit strings of length 8 either start with a 1 bit or end with the two bits 00?

Solution:

- Let A_1 be the set of strings start with '1'. By the product rule,
$$|A_1| = 2^7 = 128$$
- Let A_2 be the set of strings end with '00'. By the product rule,
$$|A_2| = 2^6 = 64$$
- $A_1 \cap A_2$ denotes the set of strings start with '1' **and** end with '00'.
By the product rule,
$$|A_1 \cap A_2| = 2^5 = 32 .$$
- By the inclusion-exclusion principle, the final answer is
$$128 + 64 - 32 = 160 .$$

Example 3 (Section 8.5)

Suppose there are 1807 students in SPMS. Of these, 453 are taking a course in Math, 567 are taking a course in Physics, and 299 are taking a course in both Math and Physics. How many are **not** taking any course in Math or Physics?



Solution: Let A_1 be the set of students taking a course in Math, and A_2 be the set of students taking a course in Physics. We have

$$|A_1| = 453, \quad |A_2| = 567, \quad \text{and} \quad |A_1 \cap A_2| = 299.$$

By the inclusion-exclusion principle, the number of students taking a course in either Math or Physics is

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2| = 453 + 567 - 299 = 721.$$

Hence, there are $1807 - 721 = 1086$ students not taking any course in Math or Physics.

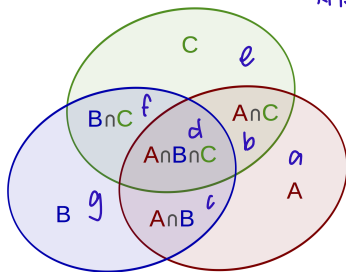
Inclusion-Exclusion Principle - Three Sets

Inclusion-Exclusion Principle

$$\begin{aligned}|A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| \\ &\quad + |A \cap B \cap C|.\end{aligned}$$

$$\text{LHS} = a + b + c + d + e + f + g$$

$$\begin{aligned}\text{RHS} &= (a + b + c + d) + (c + d + e + f) + (b + d + e + f) \\ &\quad - (c + d) - (b + d) - (d + f) + d\end{aligned}$$



Example 4 (Section 8.5)

A_1
1232 students take a course in Spanish, A_2 879 take French, A_3 114 take Russian, 103 take both Spanish and French, 23 take both Spanish and Russian, and 14 take both French and Russian. If 2092 students take at least one of these three courses, how many students take all three?

Solution: Let S, F, R be the set of students taking Spanish, French, Russian, respectively. We have $\supset A_1 \cap A_2 \cap A_3 = ?$

$$\begin{aligned} |S| &= 1232, & |F| &= 879, & |R| &= 114, \\ |S \cap F| &= 103, & |S \cap R| &= 23, & |F \cap R| &= 14, \\ |S \cup F \cup R| &= 2092 \end{aligned}$$

Insert these quantities into the inclusion-exclusion equation, we obtain

$$2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |S \cap F \cap R|$$

which gives us $|S \cap F \cap R| = 7$. Hence there are 7 students taking all three courses.

Inclusion-Exclusion Principle - General Case (Optional)

Inclusion-Exclusion Principle

usually test 2 or 3 sets only.

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \cdots \\ \cdots + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n-1} |A_1 \cap \cdots \cap A_n|.$$

or more intuitively,

$|A_1 \cup \dots \cup A_n| =$ the sum of the sizes of the individual sets
minus the sizes of all two-way intersections
plus the sizes of all three-way intersections
minus the sizes of all four-way intersections
plus the sizes of all five-way intersections
etc.

Reading Assignment

LO

Rosen (7th edition):

- Chapter 6.1, The Basics of Counting
- Chapter 8.5, Inclusion-Exclusion